



B1 Carbohydrates and lipids

B · Form and function · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Monosaccharides (e.g. glucose, fructose, galactose) join by condensation reactions, forming glycosidic bonds and releasing water, to build disaccharides and polysaccharides such as starch, glycogen and cellulose; hydrolysis reactions use water to break these bonds back down. Lipids include triglycerides, formed when glycerol condenses with three fatty acids via ester bonds, and phospholipids, which have a hydrophilic phosphate head and hydrophobic fatty acid tails; saturated fatty acids (no C=C double bonds) pack tightly and have higher melting points than unsaturated fatty acids, which have kinked chains from C=C double bonds.

Worked example: follow the reasoning

Compare the energy released from oxidising 15 g of lipid versus 15 g of carbohydrate, given lipids release approximately 39 kJ g^{-1} and carbohydrates approximately 17 kJ g^{-1} . Step 1: energy from lipid = $15 \text{ g} \times 39 \text{ kJ g}^{-1} = 585 \text{ kJ}$. Step 2: energy from carbohydrate = $15 \text{ g} \times 17 \text{ kJ g}^{-1} = 255 \text{ kJ}$. Step 3: difference = $585 - 255 = 330 \text{ kJ}$. Conclusion: the same mass of lipid releases more than double the energy of carbohydrate, explaining why lipids are an efficient long-term energy store.

Quick check · multiple choice

Which type of bond directly links a fatty acid to glycerol in a triglyceride molecule?

- A Glycosidic bond
- B Peptide bond
- C Ester bond
- D Hydrogen bond

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Starch (amylopectin) and cellulose are both polysaccharides made of glucose monomers, but they have very different biological roles. [7]
 - (a) State the type of glucose monomer (alpha or beta) found in starch and in cellulose. [1]
 - (b) Explain how differences in molecular structure between starch and cellulose relate to their different functions in plants. [4]
 - (c) Suggest why glycogen, rather than starch, is the storage polysaccharide found in animal liver and muscle cells. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Starch is made of alpha-glucose monomers; cellulose is made of beta-glucose monomers.

(b) In starch, alpha-glucose linkages allow the amylopectin chain to coil and branch, forming a compact, readily hydrolysed structure suited to short/medium-term energy storage in plant cells; in cellulose, beta-glucose linkages force each successive glucose to flip, producing straight, unbranched chains that form parallel bundles cross-linked by hydrogen bonds into strong microfibrils, giving cellulose the tensile strength needed for plant cell walls rather than being an energy store.

(c) Glycogen is even more highly branched than amylopectin, which provides many free ends for enzymes to act on simultaneously, allowing rapid hydrolysis to glucose to meet the higher and more fluctuating metabolic demands of animal cells compared with plant cells.

Quick check answer

C. A condensation reaction between the carboxyl group of a fatty acid and a hydroxyl group of glycerol forms an ester bond, releasing water.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.